

The Inverse Mostar Index Problem for Bipartite and Chemical Graphs

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Abstract

The Mostar index $\mathbf{Mo}(G)$ of a connected graph G is a distance-dependent topological descriptor that characterizes molecular graphs via numerical graph invariants. This work addresses two restricted inverse problems for the Mostar index proposed by Alizadeh et al. (2024). First, we determine exactly which nonnegative integers occur as Mostar indices of connected bipartite graphs. The answer is $\mathbb{N}_0 \setminus \{\mathbf{1}, \mathbf{3}, \mathbf{5}, \mathbf{13}\}$. Beyond the set-theoretic classification, every admissible value is realized by a connected bipartite molecular graph, that is, by a graph of maximum degree at most four. Even values follow from the known tree realization theorem; odd values are realized by explicit bipartite families, except for the nonrealizable value $\mathbf{13}$, which is excluded by a small-order analysis and a hand classification of the remaining tricyclic kernel. Second, we answer a question of Alizadeh et al. on chemical graphs by constructing an eleven-vertex chemical graph with Mostar index $\mathbf{5}$. Consequently, a nonnegative integer serves as the Mostar index of a chemical graph if and only if it is not equal to $\mathbf{1}$.

Keywords: Mostar index, inverse problem, bipartite graph, chemical graph, molecular graph, chemical tree

MSC Classification: 05C12 , 05C35 , 05C92

1 Introduction

Chemical graph theory represents a molecule by a graph whose vertices model atoms and whose edges model bonds. Numerical graph invariants, or topological indices, then

serve as compact molecular descriptors; the Wiener index is the classical distance-based example [1], and the general role of such descriptors in chemical graph theory is discussed in standard references such as [2, 3]. Within this setting, a distance-based invariant is particularly natural because it records how the molecular graph is organized through shortest-path distances.

Throughout the paper all graphs are finite, simple, connected, and undirected. For a graph G , let $V(G)$ and $E(G)$ denote its vertex set and edge set. We write $n = |V(G)|$ for the order of G , $m = |E(G)|$ for its size, $\delta(G)$ and $\Delta(G)$ for its minimum and maximum degrees, and $\text{dist}_G(x, y)$ for the ordinary shortest-path distance between vertices x and y . For any connected graph or connected multigraph H appearing below, its *cyclomatic number* is denoted by $\text{cyc}(H)$ and is defined as

$$\text{cyc}(H) = |E(H)| - |V(H)| + 1.$$

Equivalently, $\text{cyc}(H)$ is the dimension of the cycle space of H . Thus H is unicyclic, bicyclic, or tricyclic when $\text{cyc}(H) = 1, 2, 3$, respectively. A connected graph is called *chemical*, or *molecular*, if $\Delta(G) \leq 4$; a chemical tree is a chemical graph that is also a tree. We use the two terms chemical graph and molecular graph interchangeably in this paper.

For an edge $e = uv \in E(G)$, define

$$n_u(e) = |\{x \in V(G) : \text{dist}_G(x, u) < \text{dist}_G(x, v)\}|,$$

and define $n_v(e)$ analogously. The *Mostar contribution* of $e = uv$ is

$$\phi_G(e) = |n_u(e) - n_v(e)|,$$

and the *Mostar index* of G is the bond-additive invariant

$$\text{Mo}(G) = \sum_{e \in E(G)} \phi_G(e).$$

Thus each bond contributes the imbalance between the numbers of vertices that are closer to one endpoint or the other. The index was introduced by Došlić, Martinjak, Škrekovski, Tipurić Spužević, and Zubac [4]; see the survey of Ali and Došlić [5] for a broad account of subsequent results and open problems.

An integer p is said to be *realizable* by a graph invariant I over a graph class $\mathcal{G}$ if there exists a graph $G \in \mathcal{G}$ such that $I(G) = p$. Inverse problems of this kind clarify the arithmetic range of an index and reveal which values are excluded by structural restrictions such as acyclicity, bounded valency, or bipartiteness. Alizadeh, Bašić, Damnjanović, Došlić, Pisanski, Stevanović, and Xu [6] proved that a nonnegative integer is realizable as the Mostar index of a connected graph if and only if it is not equal to 1. They also initiated several restricted inverse problems. In particular, they proved that the values realized by trees are exactly the even nonnegative integers [6, Theorem 2], and asked the following question.

Question 1 (Alizadeh et al. [6]) Which nonnegative integers are Mostar indices of bipartite graphs?

They also considered chemical graphs. Their constructions realize all nonnegative integers outside $\{1, 3, 5\}$ by chemical graphs, and they observed that the value 3 is also chemically realizable. Hence the only remaining value greater than 1 was 5, leading to the following question.

Question 2 (Alizadeh et al. [6]) Is 5 the unique nonnegative integer greater than 1 that is not the Mostar index of a chemical graph?

The present paper answers both questions. Our first main theorem settles the bipartite case.

Theorem 3 *A nonnegative integer p is the Mostar index of a connected bipartite graph if and only if*

$$p \in \mathbb{N}_0 \setminus \{1, 3, 5, 13\}.$$

Moreover, every such value can be realized by a connected bipartite molecular graph.

The proof is constructive except for the single value 13. Since every tree is bipartite and the tree case is already known, all even values are covered by chemical trees. For odd values we construct three explicit unicyclic bipartite families with Mostar indices $4r - 1$, $8r + 1$, and $8r + 5$, and we treat the value 9 by a seven-vertex tricyclic graph. The values 1, 3, 5 are ruled out by parity and elementary size bounds. The value 13 is the only exceptional value not explained by parity alone; after reducing to finitely many small order-size pairs, the only nontrivial case is $(n, m) = (11, 13)$, which we exclude by a hand classification of tricyclic bipartite kernels.

Our second main theorem gives a negative answer to Question 2. In fact, the chemical graph restriction does not create any new obstruction beyond the general obstruction $p = 1$.

Theorem 4 *A nonnegative integer p is the Mostar index of a chemical graph if and only if*

$$p \in \mathbb{N}_0 \setminus \{1\}.$$

Recent work on restricted Mostar inverse problems includes inverse results for molecular unicyclic and bicyclic graphs [7], for chemical trees and unicyclic graphs [8], for tricyclic graphs and related bond-additive indices [9], and for fixed cyclo-matic number [10]. The present paper complements these works by settling the whole bipartite graph class and by completing the chemical graph case.

2 Preliminaries

For a vertex $v \in V(G)$, its *transmission* is

$$\text{Tr}_G(v) = \sum_{x \in V(G)} \text{dist}_G(v, x).$$

The following identity is useful for computations because it replaces a side-count comparison by a transmission difference.

Lemma 5 *Let G be a connected graph and let $uv \in E(G)$. Then*

$$\phi_G(uv) = |\text{Tr}_G(u) - \text{Tr}_G(v)|.$$

Proof For every vertex $x \in V(G)$, since u and v are adjacent,

$$\text{dist}_G(x, u) - \text{dist}_G(x, v) \in \{-1, 0, 1\}.$$

The vertices counted by $n_u(uv)$ contribute -1 to this difference, the vertices counted by $n_v(uv)$ contribute 1 , and the vertices equidistant from u and v contribute 0 . Therefore

$$\text{Tr}_G(u) - \text{Tr}_G(v) = \sum_{x \in V(G)} (\text{dist}_G(x, u) - \text{dist}_G(x, v)) = n_v(uv) - n_u(uv).$$

Taking absolute values gives the assertion. $\square$

We next record a parity property that is special to bipartite graphs.

Lemma 6 *Let G be a connected bipartite graph with order n and size m . Then*

$$\text{Mo}(G) \equiv nm \pmod{2}.$$

In particular, if $\text{Mo}(G)$ is odd, then both n and m are odd. In that case every edge has a positive odd contribution, and hence

$$\text{Mo}(G) \geq m.$$

Proof Let $uv \in E(G)$. Since G is bipartite and u and v lie in different parts, every vertex x has distances to u and to v of different parity. Hence no vertex is equidistant from u and v , and

$$n_u(uv) + n_v(uv) = n.$$

It follows that

$$\phi_G(uv) = |n_u(uv) - n_v(uv)| = |2n_u(uv) - n| \equiv n \pmod{2}.$$

Summing over the m edges gives $\text{Mo}(G) \equiv nm \pmod{2}$.

If $\text{Mo}(G)$ is odd, then nm is odd, so n and m are both odd. The same congruence for each edge shows that every edge contribution is odd. Because edge contributions are nonnegative, they are then at least 1 , and $\text{Mo}(G) \geq m$. $\square$

We shall also use a simple lower bound for graphs with a leaf.

Lemma 7 *Let G be a connected bipartite graph of odd order n and odd size m . If G has a leaf, then*

$$\text{Mo}(G) \geq n + m - 3.$$

Proof Let uv be a leaf edge, where u is the leaf. Removing uv separates G into components of orders 1 and $n - 1$. Thus

$$\phi_G(uv) = n - 2.$$

By Lemma 6, every other edge has contribution at least 1. Therefore

$$\text{Mo}(G) \geq (n - 2) + (m - 1) = n + m - 3.$$

□

Finally, we recall the known even case.

Theorem 8 (Alizadeh et al. [6]) *A nonnegative integer p is the Mostar index of a tree if and only if p is even. Moreover, every even nonnegative integer is realized by a chemical tree.*

Since chemical trees are bipartite molecular graphs, Theorem 8 realizes all even values needed in Theorem 3.

3 Odd values: explicit bipartite constructions

We now construct connected bipartite graphs for every odd value $p \geq 7$, except 13. In all cycle constructions, indices of vertices on C_{2r} are read modulo $2r$.

3.1 The family A_r

Let $r \geq 2$. Start with the even cycle

$$C_{2r} = v_0 v_1 \cdots v_{2r-1} v_0,$$

and add one new leaf x adjacent to v_0 . Denote the resulting graph by A_r .

Proposition 9 *For every $r \geq 2$,*

$$\text{Mo}(A_r) = 4r - 1.$$

Consequently, every odd integer $p \equiv 3 \pmod{4}$ with $p \geq 7$ is the Mostar index of a connected bipartite graph.

Proof The graph A_r has $2r + 1$ vertices and $2r + 1$ edges. Consider first an edge of the cycle C_{2r} . The $2r$ cycle vertices are split evenly by this edge: r are closer to one endpoint and r are closer to the other endpoint. The added leaf x follows its support vertex v_0 , because

$$\text{dist}_{A_r}(x, v_i) = 1 + \text{dist}_{A_r}(v_0, v_i)$$

for every cycle vertex v_i . Hence every cycle edge has contribution 1, and the total contribution of the cycle edges is $2r$.

The remaining edge v_0x is a bridge. Removing it leaves components of orders 1 and $2r$, so

$$\phi_{A_r}(v_0x) = 2r - 1.$$

Thus

$$\text{Mo}(A_r) = 2r + (2r - 1) = 4r - 1.$$

The final assertion follows by taking $r = (p + 1)/4$. $\square$

3.2 The family B_r

Let $r \geq 2$. Start with the same cycle C_{2r} . Attach a path v_0xy of length 2 at v_0 , and attach one leaf z at the opposite vertex v_r . Denote the resulting graph by B_r .

Proposition 10 *For every $r \geq 2$,*

$$\text{Mo}(B_r) = 8r + 1.$$

Consequently, every odd integer $p \equiv 1 \pmod{8}$ with $p \geq 17$ is the Mostar index of a connected bipartite graph.

Proof The three bridge edges have contributions

$$\phi_{B_r}(xy) = 2r + 1, \quad \phi_{B_r}(v_0x) = 2r - 1, \quad \phi_{B_r}(v_rz) = 2r + 1.$$

Indeed, deleting xy or v_rz separates one vertex from the remaining $2r + 2$ vertices, while deleting v_0x separates the path vertices x, y from the remaining $2r + 1$ vertices. Hence the total bridge contribution is

$$(2r + 1) + (2r - 1) + (2r + 1) = 6r + 1.$$

It remains to count the cycle edges. For every edge of C_{2r} , the cycle vertices themselves are split into two equal sets. The two added vertices x, y follow v_0 , and the vertex z follows v_r . Since v_0 and v_r are antipodal on the even cycle, they lie on opposite sides of every cycle edge. Therefore the three added vertices create an imbalance of $|2 - 1| = 1$ for each cycle edge. Thus the total contribution of the $2r$ cycle edges is $2r$.

Combining the two parts gives

$$\text{Mo}(B_r) = (6r + 1) + 2r = 8r + 1.$$

The final assertion follows by taking $r = (p - 1)/8$. $\square$

3.3 The family D_r

Let $r \geq 2$. Again start with C_{2r} and attach the path v_0xy of length 2 at v_0 . This time attach the leaf z at v_{r+1} . Denote the resulting graph by D_r .

Proposition 11 *For every $r \geq 2$,*

$$\text{Mo}(D_r) = 8r + 5.$$

Consequently, every odd integer $p \equiv 5 \pmod{8}$ with $p \geq 21$ is the Mostar index of a connected bipartite graph.

Proof The three bridge contributions are the same as in Proposition 10:

$$\phi_{D_r}(xy) = 2r + 1, \quad \phi_{D_r}(v_0x) = 2r - 1, \quad \phi_{D_r}(v_{r+1}z) = 2r + 1.$$

Their total contribution is $6r + 1$.

Now consider the cycle edges. Again, the cycle vertices are split evenly by every cycle edge. The vertices x, y follow v_0 , while z follows v_{r+1} . On the cycle C_{2r} , the vertices v_0 and v_{r+1} lie on the same side of exactly two cycle edges, namely v_0v_1 and v_rv_{r+1} , and on opposite sides of all other cycle edges. Therefore two cycle edges have contribution 3, and the remaining $2r - 2$ cycle edges have contribution 1. The total cycle contribution is

$$2 \cdot 3 + (2r - 2) \cdot 1 = 2r + 4.$$

Thus

$$\text{Mo}(D_r) = (6r + 1) + (2r + 4) = 8r + 5.$$

The final assertion follows by taking $r = (p - 5)/8$. $\square$

3.4 The value 9

It remains to realize the small odd value 9.

Proposition 12 *There is a connected bipartite graph H_9 with $\text{Mo}(H_9) = 9$.*

Proof Let H_9 be the graph obtained from

$$C_6 = v_0v_1 \cdots v_5v_0$$

by adding a new vertex x adjacent to v_0, v_2 , and v_4 . This graph is bipartite with bipartition

$$\{v_0, v_2, v_4\} \cup \{v_1, v_3, v_5, x\}.$$

It has 7 vertices and 9 edges. A direct distance computation gives

$$\text{Tr}_{H_9}(x) = 9, \quad \text{Tr}_{H_9}(v_0) = \text{Tr}_{H_9}(v_2) = \text{Tr}_{H_9}(v_4) = 10,$$

and

$$\text{Tr}_{H_9}(v_1) = \text{Tr}_{H_9}(v_3) = \text{Tr}_{H_9}(v_5) = 11.$$

Every edge of H_9 joins two vertices whose transmissions differ by 1. By Lemma 5, every edge therefore has Mostar contribution 1. Since $|E(H_9)| = 9$, we get $\text{Mo}(H_9) = 9$. $\square$

The constructions are summarized in Table 1.

Table 1 Odd bipartite realizations used in the proof.

Family	Construction	Mo	Values realized
A_r	C_{2r} with one leaf at v_0	$4r - 1$	$p \equiv 3 \pmod{4}, p \geq 7$
B_r	C_{2r} with v_0xy and a leaf at v_r	$8r + 1$	$p \equiv 1 \pmod{8}, p \geq 17$
D_r	C_{2r} with v_0xy and a leaf at v_{r+1}	$8r + 5$	$p \equiv 5 \pmod{8}, p \geq 21$
H_9	C_6 plus xv_0, xv_2, xv_4	9	$p = 9$

Corollary 13 *Every odd integer $p \geq 7$ with $p \neq 13$ is the Mostar index of a connected bipartite molecular graph.*

Proof If $p \equiv 3 \pmod{4}$, use Proposition 9. If $p \equiv 1 \pmod{8}$ and $p \geq 17$, use Proposition 10. If $p \equiv 5 \pmod{8}$ and $p \geq 21$, use Proposition 11. The only remaining odd value at least 7 is $p = 9$, which is handled by Proposition 12. All constructed graphs have maximum degree at most 3. $\square$

4 The exceptional values

We now prove that the four remaining values 1, 3, 5, 13 are not realizable by connected bipartite graphs.

Lemma 14 *No connected bipartite graph has Mostar index 1, 3, or 5.*

Proof Let G be a connected bipartite graph with $\text{Mo}(G)$ odd. By Lemma 6, n and m are odd and $\text{Mo}(G) \geq m$.

If $\text{Mo}(G) = 1$, then $m \leq 1$. The connected graphs with at most one edge are K_1 and K_2 , and both have Mostar index 0.

If $\text{Mo}(G) = 3$, then $m \leq 3$. Because n is odd, G cannot be a tree, as a tree would have $m = n - 1$ even. Thus G contains a cycle, so $m \geq n$. Hence $n \leq m \leq 3$, impossible for a bipartite graph with a cycle.

If $\text{Mo}(G) = 5$, the same reasoning gives $m \leq 5$, $m \geq n$, and n odd. The only possible case is $n = m = 5$. A connected bipartite unicyclic graph of order 5 is necessarily C_4 with one leaf attached. This is A_2 , whose Mostar index is 7 by Proposition 9. Hence 5 is not realizable. $\square$

The value 13 is slightly more delicate. We first reduce it to a finite set of small cases.

Lemma 15 *If a connected bipartite graph G satisfies $\text{Mo}(G) = 13$, then G has odd order n , odd size m , and*

$$5 \leq n \leq m \leq 13.$$

Moreover, if G has a leaf, then $n + m \leq 16$.

Proof By Lemma 6, n and m are odd and $13 = \text{Mo}(G) \geq m$, so $m \leq 13$. Since n is odd, G cannot be a tree, because then $m = n - 1$ would be even. Therefore G contains a cycle, and $m \geq n$. A bipartite graph with a cycle has at least four vertices; since n is odd, $n \geq 5$. This proves $5 \leq n \leq m \leq 13$.

If G has a leaf, Lemma 7 gives

$$13 = \text{Mo}(G) \geq n + m - 3,$$

so $n + m \leq 16$. $\square$

Lemma 16 *No connected bipartite graph G with $|V(G)| = 11$ and $|E(G)| = 13$ satisfies $\text{Mo}(G) = 13$.*

Proof Assume, to the contrary, that G is a connected bipartite graph with $n = 11$, $m = 13$, and $\text{Mo}(G) = 13$. We shall derive a contradiction by first reducing the possible structure of G , and then inspecting the resulting kernel cases.

Claim 1. Every edge of G has contribution 1, the graph G has no leaf, and the branch-degree type of G is one of

$$(5; 3), \quad (4, 3; 3), \quad (3, 3, 3; 3).$$

Here the entries before the semicolon lie in one bipartition class, and the entries after the semicolon lie in the other.

Indeed, by Lemma 6, every edge contribution is a positive odd integer. Since there are 13 edges and their sum is 13, every edge has contribution 1:

$$\phi_G(e) = 1 \quad (e \in E(G)). \quad (1)$$

In particular, G has no leaf, because a leaf edge in a graph of order 11 has contribution 9. Hence $\delta(G) \geq 2$.

Let X, Y be the bipartition of G . Since each part has degree sum 13 and every vertex has degree at least 2, the part sizes must be 5 and 6. Say $|X| = 5$ and $|Y| = 6$. The degree sequence in Y is therefore

$$3, 2, 2, 2, 2, 2,$$

and the degree sequence in X is one of

$$5, 2, 2, 2, 2; \quad 4, 3, 2, 2, 2; \quad 3, 3, 3, 2, 2.$$

Thus the vertices of degree at least 3 have exactly one of the three branch-degree types displayed in the claim.

Claim 2. The graph G is 2-connected.

Suppose that w is a cut-vertex, and let C be a component of $G - w$ with s vertices. For any neighbor u of w in C , all vertices outside C , together with w itself, are closer to w than to u . Hence

$$n_w(wu) \geq 11 - s, \quad n_u(wu) \leq s.$$

Since $\phi_G(wu) = 1$, we must have $s \geq 5$. As the components of $G - w$ have total order 10, there are exactly two components, both of order 5. Moreover, equality forces w to have exactly one neighbor in each component.

Deleting the two edges incident with w leaves two connected bipartite graphs on five vertices whose numbers of internal edges add up to 11. Hence one of them has 5 edges and the other has 6 edges. The five-vertex connected bipartite graph with 5 edges is a 4-cycle with one pendant vertex. Since G has no leaf, w must be adjacent to this pendant vertex. Let u be that pendant vertex and let a be its neighbor on the 4-cycle. For the edge ua , the vertices closer to u are u, w , and all five vertices in the other component, while the four vertices of the 4-cycle are closer to a . Thus

$$\{n_u(ua), n_a(ua)\} = \{7, 4\},$$

so $\phi_G(ua) = 3$, contradicting (1). Therefore G is 2-connected.

Claim 3. After suppressing all degree-two vertices of G , the resulting kernel K is a loopless connected multigraph of cyclomatic number 3. Each kernel edge represents an internally disjoint path in G , and the total length of these paths is 13.

Suppressing degree-two vertices preserves the cyclomatic number. Since $\text{cyc}(G) = 13 - 11 + 1 = 3$, we have

$$\text{cyc}(K) = |E(K)| - |V(K)| + 1 = 3.$$

The vertices of K are precisely the vertices of G of degree at least 3. Since G is 2-connected, K has no loop corresponding to a cycle attached at a single branch vertex. Each edge of K represents an internally disjoint path in G ; its length is odd if its endpoints are in different parts of the bipartition, and even if its endpoints are in the same part. The sum of all these path lengths is 13.

It remains to exclude the three branch-degree types from Claim 1.

Case 1: type (5; 3). Without loops, the two branch vertices would have to be joined only by parallel paths. This would give them equal degrees in the kernel, contradicting the degrees 5 and 3.

Case 2: type (4, 3; 3). Let $a, c \in X$ be the vertices of degrees 4 and 3, respectively, and let $b \in Y$ be the vertex of degree 3. If the multiplicities of the kernel edges ab , bc , and ac are x , y , and z , then

$$x + z = 4, \quad x + y = 3, \quad y + z = 3,$$

and hence $(x, y, z) = (2, 1, 2)$. Thus G consists of two internally disjoint ab -paths P_1, P_2 , one bc -path Q , and two internally disjoint ac -paths R_1, R_2 . Put

$$(p_1, p_2, q, r_1, r_2) = (\ell(P_1), \ell(P_2), \ell(Q), \ell(R_1), \ell(R_2)),$$

where $\ell(P)$ denotes the length of a path P . We may assume $p_1 \leq p_2$ and $r_1 \leq r_2$. The numbers p_1, p_2, q are odd, r_1, r_2 are even, and

$$p_1 + p_2 + q + r_1 + r_2 = 13.$$

The two paths P_1, P_2 cannot both have length 1, because G is simple. Hence the only possibilities are those listed in Table 2. For a path $P = v_0 v_1 \cdots v_s$, write $P[i]$ for the edge $v_{i-1} v_i$. In every row of the table, a direct count gives $\{n_u(e), n_v(e)\} = \{7, 4\}$ for the indicated edge $e = uv$, and so $\phi_G(e) = 3$, contradicting (1).

Table 2 The branch-degree type (4, 3; 3). Here P_1, P_2 are the ab -paths, Q is the bc -path, and R_1, R_2 are the ac -paths.

(p_1, p_2, q, r_1, r_2)	witness edge e	$\{n_u(e), n_v(e)\}$	$\phi_G(e)$
(1, 3, 1, 2, 6)	$R_2[2]$	$\{7, 4\}$	3
(1, 3, 1, 4, 4)	$R_1[2]$	$\{7, 4\}$	3
(1, 3, 3, 2, 4)	$P_2[2]$	$\{7, 4\}$	3
(1, 3, 5, 2, 2)	$Q[2]$	$\{7, 4\}$	3
(1, 5, 1, 2, 4)	$Q[1]$	$\{7, 4\}$	3
(1, 5, 3, 2, 2)	$Q[2]$	$\{7, 4\}$	3
(1, 7, 1, 2, 2)	$P_2[2]$	$\{7, 4\}$	3
(3, 3, 1, 2, 4)	$R_2[2]$	$\{7, 4\}$	3
(3, 3, 3, 2, 2)	$P_1[1]$	$\{7, 4\}$	3
(3, 5, 1, 2, 2)	$P_2[2]$	$\{7, 4\}$	3

Case 3: type (3, 3, 3; 3). Let $a, c, d \in X$ be the three branch vertices in X , and let $y \in Y$ be the branch vertex in Y . Since K has four cubic vertices and no loops, it has six kernel edges. Let the multiplicities of the three edges from y to a, c, d be μ_a, μ_c, μ_d . Then $\mu_a + \mu_c + \mu_d = 3$. If $(\mu_a, \mu_c, \mu_d) = (3, 0, 0)$, up to permutation, then the branch vertices not adjacent to y form

a separate component of the kernel, which is impossible. If $(\mu_a, \mu_c, \mu_d) = (1, 1, 1)$, the three remaining multiplicities among a, c, d are forced to be 1, 1, 1. If $(\mu_a, \mu_c, \mu_d) = (2, 1, 0)$, up to permutation, solving the three remaining degree equations gives exactly the second kernel described below. Hence there are only two 2-connected kernel types, up to permuting a, c, d .

Subcase 3.1: the simple cubic kernel. The first kernel is the simple complete graph on $\{y, a, c, d\}$: one path from y to each of a, c, d , and one path between each pair of a, c, d . Let

$$(o_a, o_c, o_d; e_{ac}, e_{ad}, e_{cd})$$

be the corresponding path lengths. The o -lengths are odd, the e -lengths are even, and their sum is 13. Up to permuting a, c, d , the six possible length patterns are exactly those in Table 3. Again, the indicated edge always has side counts $\{7, 4\}$, so every case contradicts (1).

Table 3 The simple cubic kernel on four branch vertices.

$(o_a, o_c, o_d; e_{ac}, e_{ad}, e_{cd})$	witness edge e	$\{n_u(e), n_v(e)\}$	$\phi_G(e)$
$(1, 1, 1; 2, 2, 6)$	$CD[2]$	$\{7, 4\}$	3
$(1, 1, 1; 2, 4, 4)$	$AD[3]$	$\{7, 4\}$	3
$(1, 1, 3; 2, 2, 4)$	$AC[1]$	$\{7, 4\}$	3
$(1, 1, 3; 4, 2, 2)$	$AC[2]$	$\{7, 4\}$	3
$(1, 1, 5; 2, 2, 2)$	$Yd[2]$	$\{7, 4\}$	3
$(1, 3, 3; 2, 2, 2)$	$AC[1]$	$\{7, 4\}$	3

Subcase 3.2: the second cubic kernel. The second kernel has one simple ya -path, two parallel yc -paths, two parallel ad -paths, and one simple cd -path. Let their corresponding lengths, in this order, be

$$\Lambda = (\lambda_{ya}, \lambda_{yc,1}, \lambda_{yc,2}, \lambda_{ad,1}, \lambda_{ad,2}, \lambda_{cd}).$$

We assume $\lambda_{yc,1} \leq \lambda_{yc,2}$ and $\lambda_{ad,1} \leq \lambda_{ad,2}$. The first three entries of Λ are odd, while the last three entries are even. The two parallel yc -paths cannot both have length 1. The possible length patterns with total length 13 are precisely those shown in Table 4. The indicated edge has contribution 3 in every case, another contradiction to (1).

Table 4 The second cubic kernel: paths
 $Ya, Yc_1, Yc_2, Ad_1, Ad_2, Cd$.

Λ	witness edge e	$\{n_u(e), n_v(e)\}$	$\phi_G(e)$
$(1, 1, 3, 2, 2, 4)$	$Ad_1[1]$	$\{7, 4\}$	3
$(1, 1, 3, 2, 4, 2)$	$Ad_2[2]$	$\{7, 4\}$	3
$(1, 1, 5, 2, 2, 2)$	$Cd[1]$	$\{7, 4\}$	3
$(1, 3, 3, 2, 2, 2)$	$Ad_1[1]$	$\{7, 4\}$	3
$(3, 1, 3, 2, 2, 2)$	$Ad_1[2]$	$\{7, 4\}$	3

All possible kernels and all possible length assignments have now been covered. Each case contains an edge of contribution 3, contradicting (1). Therefore no such graph G exists. $\square$

Lemma 17 *No connected bipartite graph has Mostar index 13.*

Proof Assume that G is a connected bipartite graph with $\text{Mo}(G) = 13$. By Lemma 15, G has odd order and odd size with $5 \leq n \leq m \leq 13$. Since a simple bipartite graph of order n has at most $\lfloor n^2/4 \rfloor$ edges, the only feasible pairs are

$$(5, 5), (7, 7), (7, 9), (7, 11), (9, 9), (9, 11), (9, 13), (11, 11), (11, 13), (13, 13).$$

We exclude these pairs case by case, without enumerating labelled graphs.

Case 1: the unicyclic pairs $m = n$. If $n \geq 9$, then every bipartite unicyclic graph of odd order has a leaf, and Lemma 7 gives

$$\text{Mo}(G) \geq n + m - 3 = 2n - 3 \geq 15.$$

Thus $(9, 9), (11, 11), (13, 13)$ cannot occur. For $(5, 5)$, the graph is necessarily $K_{2,3}$ with one edge deleted, equivalently a 4-cycle with a leaf attached; its edge contributions are 3, 1, 1, 1, 1, so $\text{Mo}(G) = 7$. For $(7, 7)$, the unique cycle has length 4 or 6. If it has length 6, then the graph is A_3 , and Proposition 9 gives $\text{Mo}(G) = 11$. If the unique cycle has length 4, the three remaining vertices form a rooted forest attached to the cycle. When this forest has at least two leaves, two leaf edges already contribute $5 + 5$, and the other five edges contribute at least 1 each, so $\text{Mo}(G) \geq 15$. When it has exactly one leaf, those three vertices form a path of length three attached at a cycle vertex; the three path edges have contributions 1, 3, 5, and the four cycle edges have contribution 3 each, giving $\text{Mo}(G) = 21$. Hence no unicyclic feasible pair gives 13.

Case 2: $(n, m) = (7, 9)$. First suppose that G has a leaf. Remove the leaf and call the remaining graph H . Then H has six vertices and eight edges. According to its bipartition, H is either $K_{2,4}$ or $K_{3,3}$ with one edge deleted. In the first case, if the deleted leaf is attached at one of the two vertices of the part of size two, then every edge from this support vertex to the part of size four has side counts $\{5, 2\}$ in G ; if the leaf is attached in the part of size four, then every edge from either vertex of the part of size two to a different vertex of the part of size four has side counts $\{5, 2\}$. In the second case, write $H = K_{3,3} - xy$. Up to symmetry there are four positions for the leaf support: x, y , a vertex in the part of x different from x , or a vertex in the part of y different from y . In each of these four positions, one immediately finds an edge with side counts $\{5, 2\}$: respectively an edge from $X \setminus \{x\}$ to y , an edge from x to $Y \setminus \{y\}$, an edge from x to $Y \setminus \{y\}$, or an edge from $X \setminus \{x\}$ to y . Thus a leaf-containing graph with $(n, m) = (7, 9)$ has a leaf edge of contribution 5 and another edge of contribution at least 3. The remaining seven edges have positive odd contribution, and hence $\text{Mo}(G) \geq 5 + 3 + 7 = 15$.

Now suppose that G has no leaf. Then its bipartition has sizes 3 and 4. Let the degree sequence in the part of size three be nondecreasing. If it is $(3, 3, 3)$, then the degree sequence in the other part is $(3, 2, 2, 2)$, and G is $K_{3,4}$ with a matching of size three deleted. A direct distance count on this unique structure shows that all nine edge contributions are 1, so $\text{Mo}(G) = 9$. If the degree sequence in the part of size three is $(2, 3, 4)$, then, up to isomorphism, the missing edges of $K_{3,4}$ are incident with the vertices of degrees 2 and 3. The degree-four vertex is adjacent to three degree-two vertices in the other part. For each such incident edge, the endpoint of degree four, its other three neighbors, and the remaining vertex in the same part are all closer to the degree-four endpoint; hence the edge has side counts $\{5, 2\}$. There are at least three such edges, and therefore $\text{Mo}(G) \geq 9 + 3 \cdot 2 = 15$. Thus $(7, 9)$ cannot realize 13.

Case 3: $(n, m) = (7, 11)$. The bipartition must have sizes 3 and 4, and $G = K_{3,4} - xy$ for some edge xy . The two vertices in the part of size three different from x are both adjacent to y ; the two edges joining them to y have side counts $\{5, 2\}$. Hence at least two edges have contribution 3, while all other edges have positive odd contribution. Therefore $\text{Mo}(G) \geq 11 + 2 \cdot 2 = 15$.

Case 4: $(n, m) = (9, 11)$. The leaf bound excludes leaves, so $\delta(G) \geq 2$. The bipartition must have sizes 4 and 5, because a part of size six would already contribute degree sum at least $12 > 11$. Suppressing all degree-two vertices gives a tricyclic bipartite kernel of the same three branch-degree types as in Lemma 16. The type $(5; 3)$ is impossible without loops, as before. The remaining length patterns have total length 11, and they are listed in Table 5. The last column gives the exact lower bound obtained by elementary distance counts along the displayed kernel paths. In every case it is greater than 13.

Table 5 The leafless case $(n, m) = (9, 11)$. Path lengths have the same notation as in Lemma 16; permutations that only relabel symmetric branch vertices are suppressed.

Kernel type	possible length patterns	lower bound
$(4, 3; 3)$	$(1, 3, 1, 2, 4), (1, 3, 3, 2, 2), (1, 5, 1, 2, 2), (3, 3, 1, 2, 2)$	$\text{Mo}(G) \geq 23$
Simple cubic	$(1, 1, 1; 2, 2, 4), (1, 1, 3; 2, 2, 2)$	$\text{Mo}(G) \geq 19$
Second cubic	$(1, 1, 3, 2, 2, 2)$	$\text{Mo}(G) = 23$

Case 5: $(n, m) = (9, 13)$. Again the leaf bound gives $\delta(G) \geq 2$. We show that some edge has contribution at least 3, which is impossible because Lemma 6 would force all thirteen edge contributions to be exactly 1.

If the bipartition has sizes 3 and 6, choose a vertex x in the part of size three with maximum degree. Since the degree sum on this part is 13, we have $\deg(x) \geq 5$. The part of size six has five vertices of degree two and one of degree three, so x is adjacent to some degree-two vertex y . For the edge xy , the vertex x and the other neighbors of x already account for at least five vertices closer to x . The degree-two vertex y has exactly one neighbor in the part of x besides x ; let z be the third vertex in that part. Since z has degree at least two and x misses at most one vertex in the opposite part, z shares a neighbor with x . Hence $d(z, x) = 2 < d(z, y)$, and z is also closer to x . Thus $n_x(xy) \geq 6$, so $\phi_G(xy) \geq 3$.

If the bipartition has sizes 4 and 5, choose a vertex x of maximum degree in the part of size four. Then $\deg(x) \geq 4$. In the part of size five there are at least two degree-two vertices. Since x has at least four neighbors, it is adjacent to some degree-two vertex y . For the edge xy , the vertex x and the other $\deg(x) - 1$ neighbors of x are closer to x . The vertex y has exactly one neighbor in the part of size four besides x ; the other two vertices of that part are not adjacent to y . Each of these two vertices has degree at least two, while x misses at most one vertex in the opposite part, so each shares a neighbor with x and is closer to x than to y . Thus $n_x(xy) \geq 6$, and again $\phi_G(xy) \geq 3$. This contradiction excludes $(9, 13)$.

Case 6: $(n, m) = (11, 13)$. This is exactly the exceptional small case excluded by Lemma 16.

All feasible pairs have now been excluded. Therefore no connected bipartite graph has Mostar index 13. $\square$

We can now prove the main theorem.

Proof of Theorem 3 Every even nonnegative integer is the Mostar index of a chemical tree by Theorem 8. Since every chemical tree is a bipartite molecular graph, all even values are realized in the required class.

Every odd integer $p \geq 7$, except 13, is realized by a connected bipartite molecular graph by Corollary 13. Finally, Lemmas 14 and 17 show that 1, 3, 5, 13 are not realizable by any connected bipartite graph. This proves the theorem. $\square$

5 Chemical graphs and the value 5

In this section we complete the inverse problem for chemical graphs. The only ingredient missing from the constructions of Alizadeh et al. [6] is a chemical graph with Mostar index 5. We now give such a graph explicitly.

Let G_5 be the graph with vertex set

$$\{a_0, a_1, a_2, a_3, a_4, z, b_0, b_1, b_2, b_3, b_4\}.$$

Its edge set consists of the following nineteen edges:

$$a_0a_1, a_1a_2, a_2a_3, a_3a_4, a_4a_0,$$

$$za_0, za_2, za_3, za_4,$$

$$b_0b_1, b_1b_2, b_2b_3, b_3b_4, b_4b_0,$$

$$a_0b_0, a_4b_1, a_1b_2, a_3b_3, a_2b_4.$$

Figure 1 illustrates the construction.

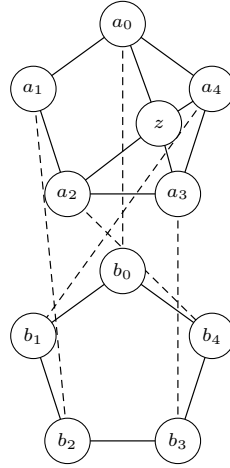


Fig. 1 The chemical graph G_5 . The dashed edges are the matching $a_0b_0, a_4b_1, a_1b_2, a_3b_3, a_2b_4$ between the two 5-cycles.

Thus the vertices a_0, a_1, a_2, a_3, a_4 form a 5-cycle, the vertices b_0, b_1, b_2, b_3, b_4 form another 5-cycle, the vertex z is adjacent to a_0, a_2, a_3, a_4 , and the two cycles are joined by the indicated matching.

Proposition 18 *The graph G_5 is a chemical graph and $\text{Mo}(G_5) = 5$.*

Proof The degrees are

$$\deg(a_0) = 4, \quad \deg(a_1) = 3, \quad \deg(a_2) = \deg(a_3) = \deg(a_4) = 4, \quad \deg(z) = 4,$$

while

$$\deg(b_0) = \deg(b_1) = \deg(b_2) = \deg(b_3) = \deg(b_4) = 3.$$

Hence $\Delta(G_5) = 4$, so G_5 is chemical.

A direct distance count gives the transmission values in Table 6. The table records, for each type of vertex, the numbers (N_0, N_1, N_2, N_3) , where N_i is the number of vertices at distance i from the chosen vertex.

Table 6 Distance distributions and transmissions in G_5 .

Vertices	Distance distribution (N_0, N_1, N_2, N_3)	Transmission
a_0, a_2, a_3, a_4, z	$(1, 4, 5, 1)$	17
a_1	$(1, 3, 7, 0)$	17
b_0, b_1, b_2, b_3, b_4	$(1, 3, 6, 1)$	18

Therefore

$$\text{Tr}_{G_5}(a_0) = \text{Tr}_{G_5}(a_1) = \cdots = \text{Tr}_{G_5}(a_4) = \text{Tr}_{G_5}(z) = 17$$

and

$$\text{Tr}_{G_5}(b_0) = \text{Tr}_{G_5}(b_1) = \cdots = \text{Tr}_{G_5}(b_4) = 18.$$

By Lemma 5, every edge whose endpoints have the same transmission has Mostar contribution 0, while every edge joining a vertex of transmission 17 to a vertex of transmission 18 has Mostar contribution 1. The only such edges are

$$a_0b_0, \quad a_4b_1, \quad a_1b_2, \quad a_3b_3, \quad a_2b_4.$$

Consequently,

$$\text{Mo}(G_5) = 5.$$

□

We can now prove the chemical graph classification.

Proof of Theorem 4 Alizadeh et al. [6] proved that no connected graph has Mostar index 1, so no chemical graph has Mostar index 1. They also constructed chemical graphs realizing every nonnegative integer outside $\{1, 3, 5\}$, and they gave a chemical graph realizing 3. Proposition 18 supplies the remaining value 5. Hence every nonnegative integer except 1 is realized by a chemical graph, and 1 is not realized by any chemical graph. □

6 Concluding remarks

We have solved two restricted inverse problems for the Mostar index. First,

$$\{\text{Mo}(G) : G \text{ is connected and bipartite}\} = \mathbb{N}_0 \setminus \{1, 3, 5, 13\}.$$

This result is slightly stronger than the statement of the restricted inverse problem, since every realizable value can be attained by a bipartite molecular graph. Apart from the isolated value 9, all odd bipartite realizations in this paper are unicyclic. Thus

the remaining exceptional value 13 is not caused by parity alone, but by a small-order obstruction.

Second,

$$\{\text{Mo}(G) : G \text{ is a chemical graph}\} = \mathbb{N}_0 \setminus \{1\}.$$

The graph G_5 constructed in Section 5 shows that the value 5 is chemically realizable, so the chemical graph case has the same range as the unrestricted connected-graph case.

Two related directions remain natural. First, one may ask for analogous exact inverse problems in more restrictive bipartite subclasses, for example bipartite cactus graphs. Second, one may study minimal order and minimal size among graphs in a prescribed class realizing a fixed admissible Mostar index.

Statements and Declarations

Funding. This work was supported by the Natural Science Foundation of Shandong Province (Grant No. ZR2024MA073).

Competing interests. The authors declare that they have no competing interests.

Author contributions. Both authors contributed to the study conception, mathematical arguments, and manuscript preparation. Both authors read and approved the final manuscript.

Data availability. No datasets were generated or analyzed during the current theoretical study.

Code availability. No code is needed for the proofs in this paper. The chemical graph G_5 is specified by its edge list in Section 5.

Ethics approval and consent to participate. Not applicable.

Conflict of interest. The authors declare that they have no known competing financial interests or personal relationship that could have appeared to influence the work reported in this paper.

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